

Selectime: Validation-selected inputs for Chronos-2

1 Candidate family

Given an observed target prefix and a horizon H , compare independent univariate forecasting, joint multivariate forecasting, past-only target augmentation with $\sqrt{|x|}$ and $\text{sign}(x)$, and retrieved covariates for $K \in \{1, 5, 10, 15\}$. The backbone is frozen Chronos-2 and every candidate produces a median point forecast. There is no parameter fitting.

2 Aligned retrieval

For a fixed retrieval lookback r , represent each query and neighbor as $z(x) = (x - \mu(x)) / \max(\sigma(x), 10^{-8})$, using only that lookback’s finite past values. Rank eligible neighbors by Euclidean distance on finite overlap (at least 80%, with squared distances adjusted for overlap). Require a complete maximum- K list; smaller K values reuse prefixes. The datastore is within one dataset/frequency task, across items and variates by default. A same-series ablation is retained. Task-specific calendar phases and frequency-dependent strides define the eligible dates.

Before supplying a neighbor trajectory s_i to the backbone, apply

$$\tilde{s}_i = (s_i - \mu_i) \frac{\max(\sigma_q, 10^{-8})}{\max(\sigma_i, 10^{-8})} + \mu_q,$$

where both sets of statistics use past retrieval lookbacks only. Supply the rescaled neighbor past and observed future as covariates. Self-augmentation has no future covariates.

3 Chronology and selection

The validation datastore ends before validation starts. The validation period immediately precedes the official TIME test interval. After freezing method and K , expand retrieval to all eligible pre-test history. Complete neighbor futures must end before the applicable boundary and be observed before the query; no test observation enters the datastore.

Score candidates by mean squared error divided by the mean seasonal squared difference over the observed target prefix (MSSE). Average variates within each validation date before comparing dates for task-level selection; per-variate selection scores each item/variate’s dates independently. Use Adaptime’s paired moving-date-block bootstrap: a candidate is admissible when its mean loss difference from the observed best is at most one bootstrap standard error. Prefer univariate, multivariate, self-augmentation, then smaller K within that threshold. Choices remain fixed throughout testing.

4 Evidence

Official test windows and the Seasonal Naive metric-cell grid remain unchanged. Unavailable retrieval and non-finite candidate cells use canonical univariate forecasts with recorded masks. No Selectime results have been analyzed.